DASC 3240, L14

PCA and others – Assignments

2023/24 SPRING SEMESTER

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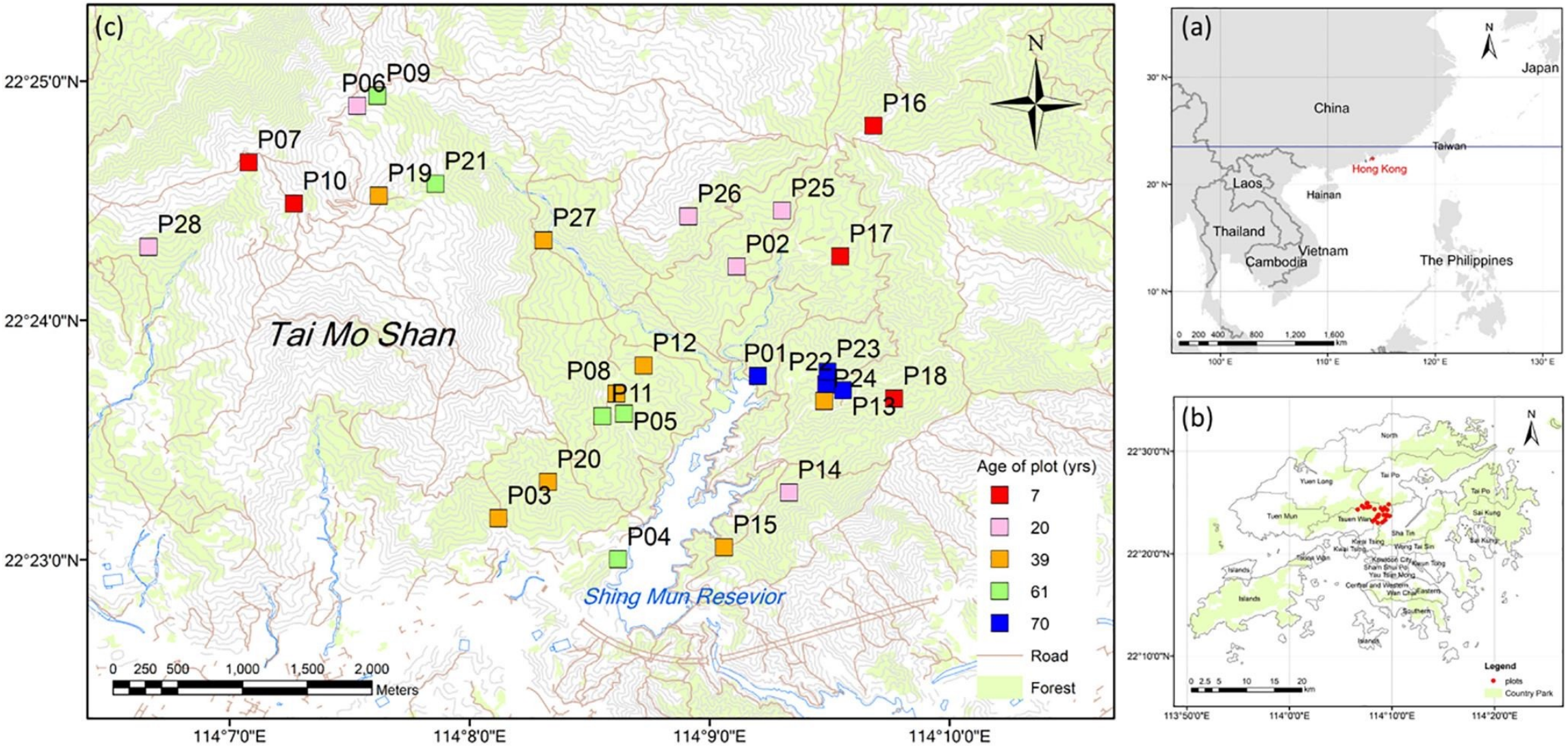
Data

Tree community data in Hong Kong

Kadoorie Farm & Botanical Garden

https://www.kfbg.org/en/flora-conservation/forest-dynamic-plot

Tree community data in Hong Kong



Abbas et al. (2019) *Ecological Indicators* https://doi.org/10.1016/j.ecolind.2019.105524

Tree community data in Hong Kong

Total 28 forest plots
Each plot is 20 m × 20 m (total = 1.12 ha)
Total 8,575 individuals, 229 species, 63 families

Table 1. Number of plots under each age class.

Age of succession	Number of plots	Age midpoint
(0–14]	5	7
(14 – 26]	6	20
(26 – 52]	8	39
(52–70]	5	61
>70	4	70

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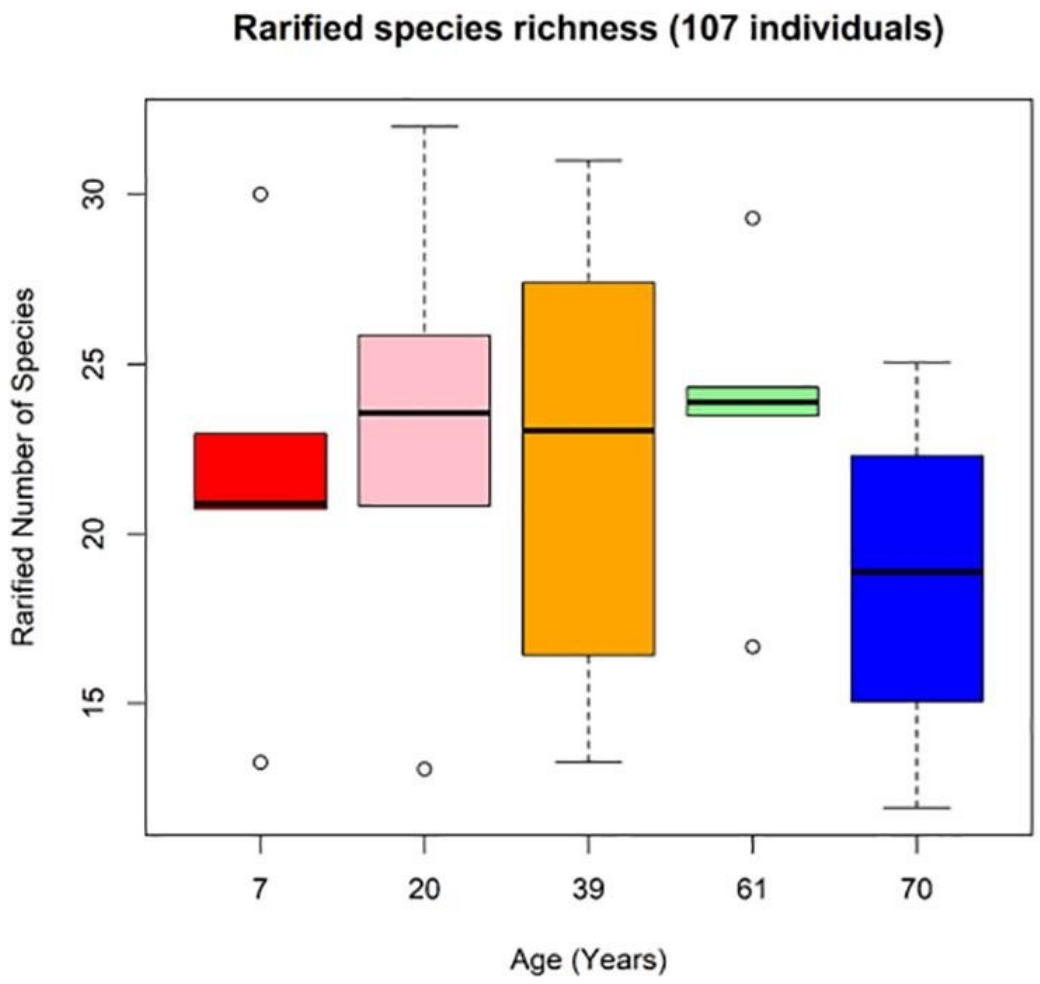
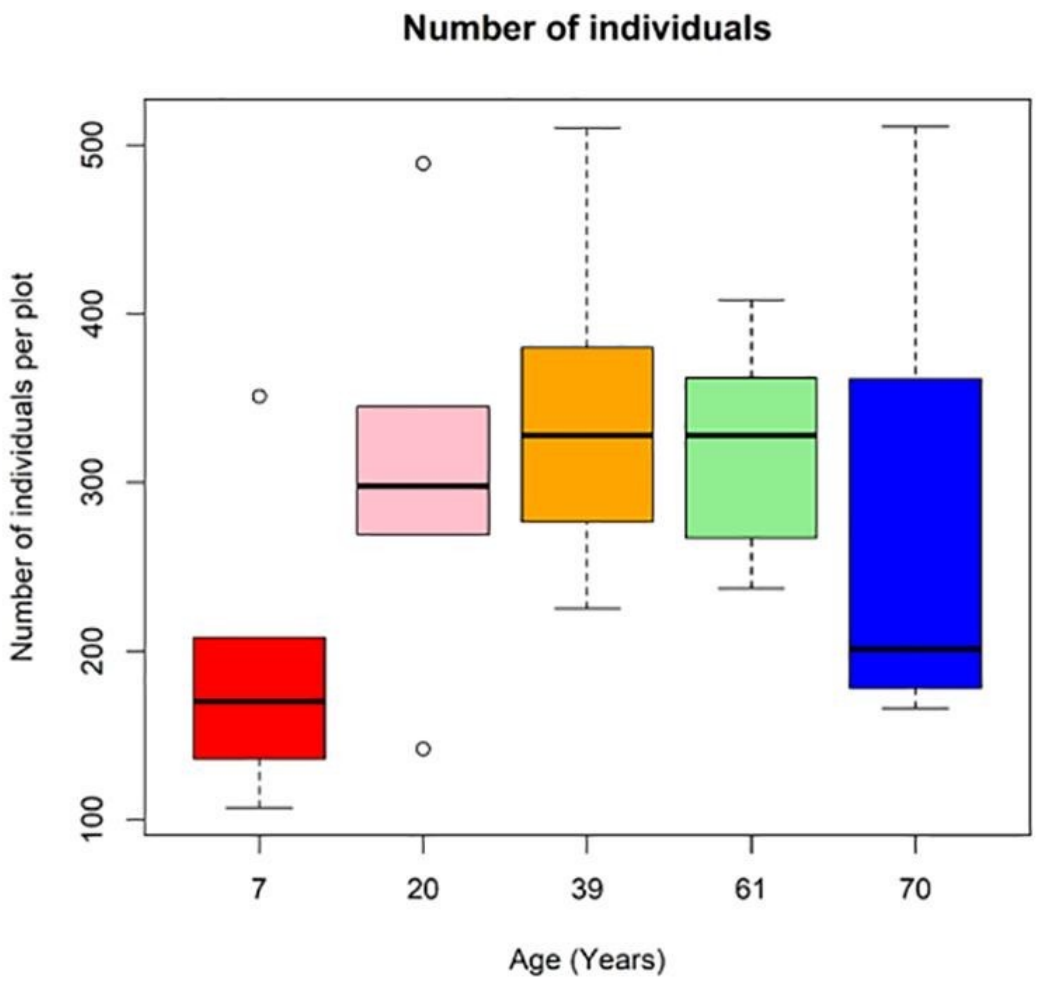
Tree community data in Hong Kong





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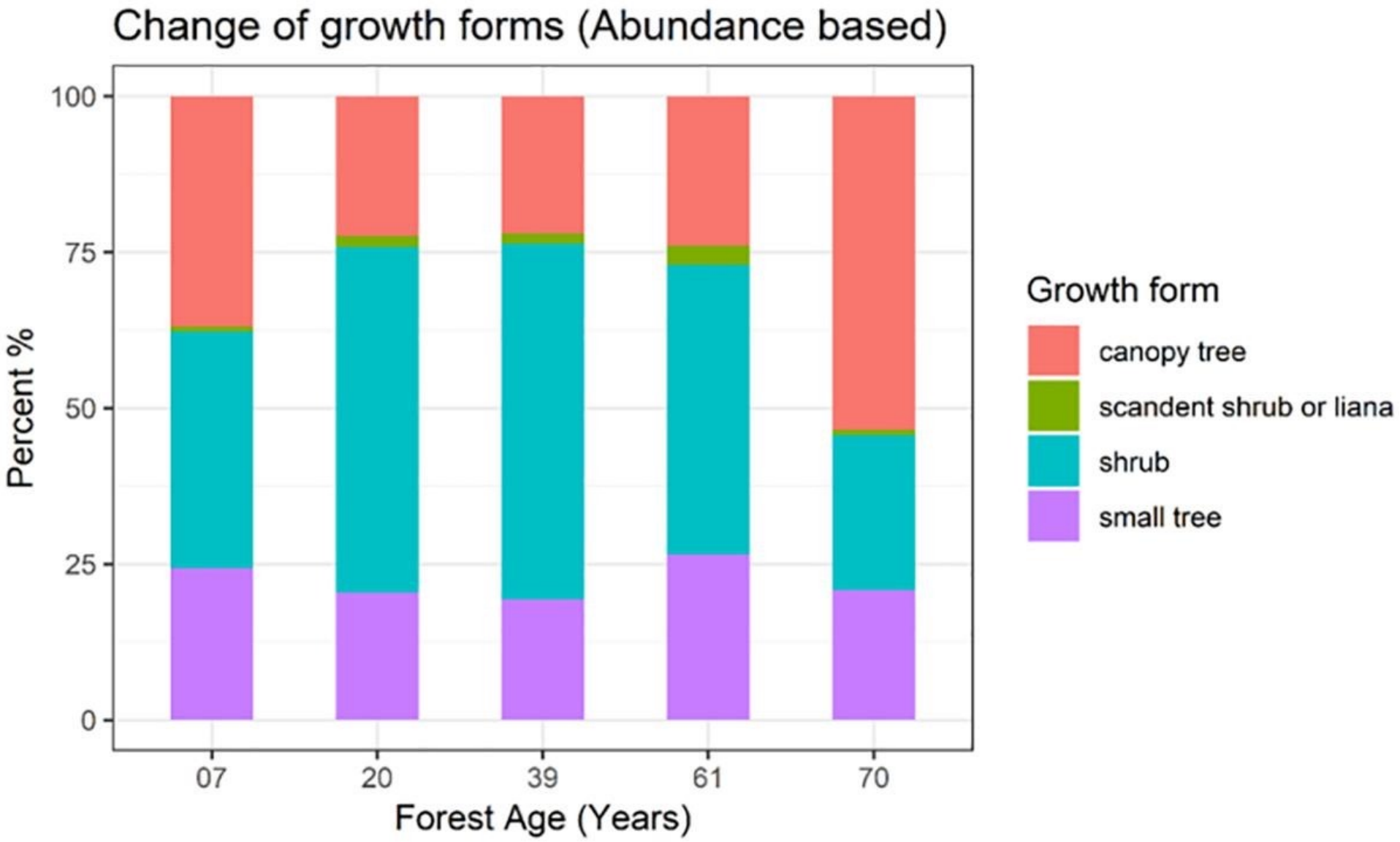
Tree community data in Hong Kong



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Assignments

DIMENSION REDUCTION

Assignments

- **Tree community data in Hong Kong**

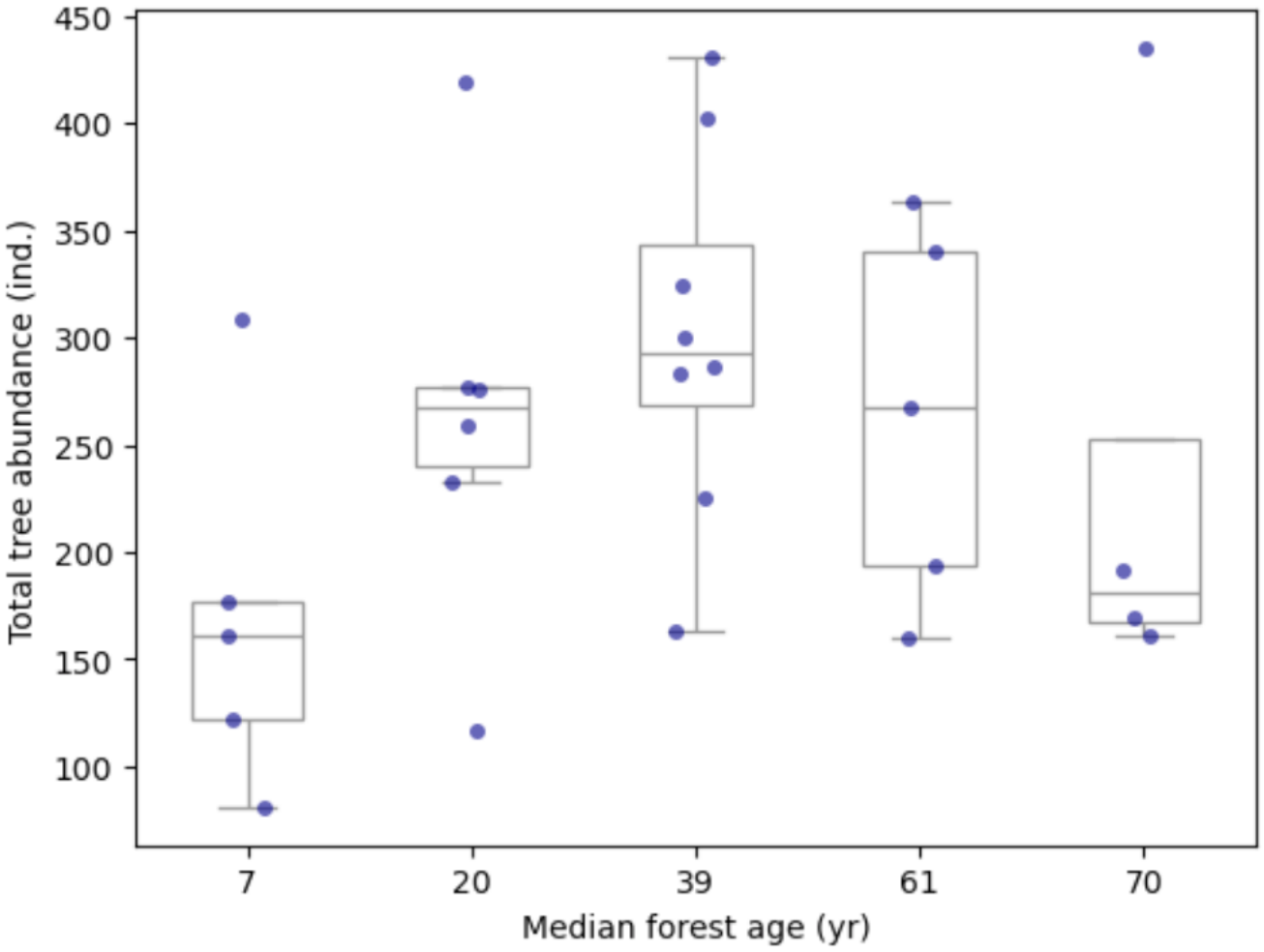
  Go to https://github.com/ong8181/DASC3240/blob/main/data/ and download "`Abbas_2019_TableS4_ed1.csv`" and "`Abbas_2019_TableS4_ed2.csv`".
  - "`Abbas_2019_TableS4_ed1.csv`" includes the plot information.
  - "`Abbas_2019_TableS4_ed2.csv`" includes dominant tree species data in the 28 forest plots.

  1. Calculate the total abundance of the dominant tree species in each forest plot, and visualize the relationship between the total abundance and forest age. You should use "AGE_MEDIAN" column in the "`Abbas_2019_TableS4_ed1.csv`" file as a forest age index.
  2. Visualize the dominant tree species compositions using PCA, NMDS, and t-SNE.

  [**Optional task**] Discuss the effects of forest age ("AGE_MEDIAN") on the tree species compositions based on your visualization in Tasks 1 and 2.

Examples

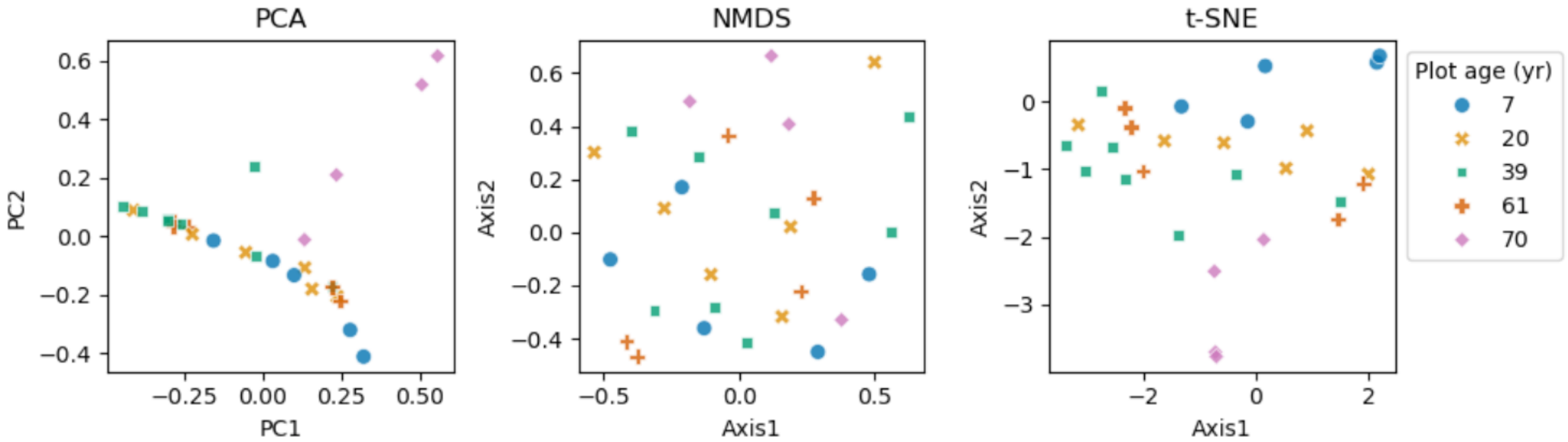
- **Tree community data in Hong Kong**
  1. Calculate the total abundance of the dominant tree species in each forest plot, and visualize the relationship between the total abundance and forest age. You should use “AGE_MEDIAN” column in the “`Abbas_2019_TableS4_ed1.csv`” file as a forest age index.



Examples

- **Tree community data in Hong Kong**

  2. Visualize the dominant tree species compositions using PCA, NMDS, and t-SNE.



You might get different figures as NMDS and t-SNE involve random processes (I used 1234 as a random seed, btw).

Notes

- **You should consider the following points**:
  1. Please remember to add appropriate axis labels (and units, if appropriate).
  2. Set random seeds if your analysis includes random process(es).
  3. Before applying dimension reduction functions (e.g., `PCA()`), please make sure that rows represent samples (= forest plots) and columns represent tree species.
  4. Usually we should consider whether we must perform the data preprocessing before applying dimension reduction techniques (e.g., standardization). In this assignment, please convert your data compositional using the following code `df = df.div(df.sum(axis = 0), axis = 1)` (`df` represents the name of your pandas dataframe. You should change it depending on your object name).